Data Sources Reference Document

Korea Petrochemical Net Zero Pathway Analysis

PLANiT Institute
Seoul, Republic of Korea

Technical Documentation

Version 1.0

January 2026

Color Legend

Green = Properly cited source
Yellow = Partial source (needs specific reference)
Red = No source documented

Document Purpose:
This document provides a comprehensive audit of all data sources used in the Korea Petrochemical Net Zero Pathway Analysis model. Values without proper academic or industry citations are highlighted in **red**.

Contents

1 Executive Summary

This document audits the data sources for all parameters used in the Korea Petrochemical MACC model. The analysis reveals:

- **Well-documented:** Emission factors, renewable energy prices, and core technology specifications
- **Undocumented:** Product energy benchmarks, fuel prices, emission targets, and asset valuation parameters

1.1 Summary Statistics

Category	Values	With Source	Status
Emission Factors	9	9	Complete
Technology Parameters	25	18	Partial [PARTIAL]
RE Price Trajectory	26	26	Complete
H2 Price Trajectory	26	26	Derived
Product Benchmarks	432	0	None [SOURCE NEEDED]
Fuel Prices	182	0	None [SOURCE NEEDED]
Emission Targets	6	0	None [SOURCE NEEDED]
Asset Valuation	12	0	None [SOURCE NEEDED]
Total	**718**	**79**	**11%**

Table 1: Data source coverage summary

2 Technology Parameters

Source File: `data/assumptions/technology_parameters.csv`

2.1 Heat Pump

Parameter	Value	Source	Status
COP (Coefficient of Performance)	4.0	Kosmadakis (2020)	OK
Energy Conversion Efficiency	95%	Internal [SOURCE NEEDED]	Missing [SOU
Application Temperature	<165°C	Kosmadakis (2020)	OK
CAPEX 2025	\$800 M/MtCO_2	Internal [SOURCE NEEDED]	Missing [SOU
CAPEX 2050	\$400 M/MtCO_2	Internal [SOURCE NEEDED]	Missing [SOU
CAPEX Learning Rate	50% by 2050	Internal [SOURCE NEEDED]	Missing [SOU
OPEX	3% of CAPEX	Internal [SOURCE NEEDED]	Missing [SOU
Lifetime	20 years	Internal [SOURCE NEEDED]	Missing [SOU
TRL	9	Internal [SOURCE NEEDED]	Missing [SOU
Available Year	2025	Internal [SOURCE NEEDED]	Missing [SOU

Table 2: Heat Pump technology parameters

2.2 NCC-H2 (Green Hydrogen Furnaces)

Parameter	Value	Source	Status
H2 Consumption	0.2 t-H2/t-C2H4	Power Engineering 2022	OK
H2 Demand Factor	0.127 t/tCO_2	Derived [PARTIAL]	Partial [PARTIAL]
Energy Efficiency	85%	Power Engineering 2022	OK
Efficiency Basis	HHV/LHV conservative	Coolbrook (2024)	OK
CAPEX 2025	\$1,700 M/MtCO_2	Internal [SOURCE NEEDED]	Missing [SOURCE N
CAPEX 2050	\$850 M/MtCO_2	Internal [SOURCE NEEDED]	Missing [SOURCE N
OPEX	4% of CAPEX	Internal [SOURCE NEEDED]	Missing [SOURCE N
Lifetime	25 years	Internal [SOURCE NEEDED]	Missing [SOURCE N
TRL	7	Internal [SOURCE NEEDED]	Missing [SOURCE N
Available Year	2030	Internal [SOURCE NEEDED]	Missing [SOURCE N

Table 3: NCC-H2 technology parameters

2.3 NCC-Electricity (Electric Crackers)

Parameter	Value	Source	Status
Electricity Consumption	5.0 MWh/t-C2H4	BASF et al. (2024)	OK
Electricity Demand Factor	3.18 MWh/tCO_2	Derived [PARTIAL]	Partial [PARTIAL]
Energy Efficiency	95%	BASF et al. (2024)	OK
CAPEX 2025	$1,500 M/Mt$CO_2$	Internal [SOURCE NEEDED]	Missing [SOURCE
CAPEX 2050	$750 M/Mt$CO_2$	Internal [SOURCE NEEDED]	Missing [SOURCE
OPEX	4% of CAPEX	Internal [SOURCE NEEDED]	Missing [SOURCE
Lifetime	25 years	Internal [SOURCE NEEDED]	Missing [SOURCE
TRL	8	Internal [SOURCE NEEDED]	Missing [SOURCE
Available Year	2030	Internal [SOURCE NEEDED]	Missing [SOURCE

Table 4: NCC-Electricity technology parameters

2.4 RDH (RotoDynamic Heater)

Parameter	Value	Source	Status
Energy Efficiency	93%	Coolbrook (2024)	OK
Application	High-temp BTX	Coolbrook (2024)	OK
CAPEX 2025	$900 M/Mt$CO_2$	Internal [SOURCE NEEDED]	Missing [SOURCE NEEDED]
CAPEX 2050	$450 M/Mt$CO_2$	Internal [SOURCE NEEDED]	Missing [SOURCE NEEDED]
OPEX	3% of CAPEX	Internal [SOURCE NEEDED]	Missing [SOURCE NEEDED]
Lifetime	25 years	Internal [SOURCE NEEDED]	Missing [SOURCE NEEDED]
TRL	8	Internal [SOURCE NEEDED]	Missing [SOURCE NEEDED]
Available Year	2026	Internal [SOURCE NEEDED]	Missing [SOURCE NEEDED]

Table 5: RDH technology parameters

3 Emission Factors

Source File: `data/assumptions/emission_factors.csv`

Fuel	Value	Unit	Source
Naphtha	0.0542	tCO_2/GJ	Intergovernmental Panel on Climate Change (2019),
LNG	0.0561	tCO_2/GJ	Intergovernmental Panel on Climate Change (2019),
Fuel Gas	0.050	tCO_2/GJ	American Petroleum Institute (2021)
Byproduct Gas	0.048	tCO_2/GJ	American Petroleum Institute (2021)
LPG	0.0631	tCO_2/GJ	Intergovernmental Panel on Climate Change (2019),
Fuel Oil	0.0773	tCO_2/GJ	Intergovernmental Panel on Climate Change (2019),
Diesel	0.0741	tCO_2/GJ	Intergovernmental Panel on Climate Change (2019),
Grid Electricity (2025)	0.436	tCO_2/MWh	Korea MOE 2024 [PARTIAL]
Green H2	0.0	tCO_2/kg	Electrolysis assumption

Table 6: Emission factors by fuel type

Notes:

- LNG and Fuel Gas values were corrected from 0.0149 in earlier versions
- Grid electricity emission factor references "Korea 2025" but lacks specific MOE document number

4 Energy Price Trajectories

4.1 Renewable Electricity Price (RE-PPA)

Source File: `data/assumptions/prices/re_price_trajectory.csv`

Year	Price ($/MWh)	Source
2025	65.00	International Renewable Energy Agency (2024), International Energy Ag
2030	55.69	International Renewable Energy Agency (2024), International Energy Ag
2035	47.71	International Renewable Energy Agency (2024), International Energy Ag
2040	40.87	International Renewable Energy Agency (2024), International Energy Ag
2045	35.02	International Renewable Energy Agency (2024), International Energy Ag
2050	30.00	International Renewable Energy Agency (2024), International Energy Ag

Table 7: Renewable electricity price trajectory (selected years)

Note: Prices for all 26 years (2025-2050) are interpolated from anchor years. Source: Korea RE PPA trajectory based on IRENA 2024 and IEA WEO 2024.

4.2 Green Hydrogen Price (LCOH)

Source File: `data/assumptions/prices/h2_price_trajectory.csv`

Year	Price ($/kg)	RE Price	Ely CAPEX	Efficiency
2025	4.58	$65/MWh	$1,000/kW	70%
2030	3.91	$55.7/MWh	$900/kW	71%
2035	3.33	$47.7/MWh	$800/kW	72%
2040	2.82	$40.9/MWh	$700/kW	73%
2045	2.39	$35.0/MWh	$600/kW	74%
2050	2.01	$30.0/MWh	$500/kW	75%

Table 8: Green hydrogen price trajectory with underlying assumptions

Source: PLANiT LCOH Model (internal calculation based on cited RE prices)

LCOH Formula:

$$\text{LCOH} = \frac{\text{CAPEX} \times \text{CRF} + \text{Fixed OPEX} + \text{Stack Replacement}}{\text{Annual H}_2\text{ Production}} + \text{Electricity Cost} \quad (1)$$

Where CRF = Capital Recovery Factor (8% discount rate, 20-year lifetime)

4.3 Fuel Prices

Source File: `data/assumptions/prices/fuel_price_trajectory.csv`

Fuel	Price (2025-2050)	Source	Status
Naphtha	$15.0/GJ (constant) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing
LNG	$12.0/GJ (constant) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing
Fuel Gas	$10.0/GJ (constant) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing
LPG	$14.0/GJ (constant) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing
Fuel Oil	$13.0/GJ (constant) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing
Diesel	$16.0/GJ (constant) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing
Grid Electricity	$0.10/kWh (constant) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing

Table 9: Fuel price assumptions – ALL UNDOCUMENTED

Critical Issue: All fuel prices are constant from 2025-2050 with no trajectory or justification. Recommendation: Update with IEA or EIA projections.

4.4 Grid Emission Factor Trajectory

Source File: `data/assumptions/prices/grid_emission_trajectory.csv`

Year	Grid EF (tCO_2/MWh)	Source	Status
2025	0.436	Korea Grid 2025 [PARTIAL]	Partial [PARTIAL]
2030	0.349	Grid pathway [PARTIAL]	Partial [PARTIAL]
2040	0.140	Grid pathway [PARTIAL]	Partial [PARTIAL]
2050	0.000	Net Zero assumption	OK

Table 10: Grid emission factor trajectory

Issue: References "Korea 2025" without specific MOE document number or publication.

4.5 Grid Price Trajectory

Source File: `data/assumptions/prices/grid_price_trajectory.csv`

Year	Price ($/MWh)	Source	Status
2025	$100.00 [SOURCE NEEDED]	User-specified [SOURCE NEEDED]	Missing [SOURCE NEE
2030	$118.28 [SOURCE NEEDED]	User-specified [SOURCE NEEDED]	Missing [SOURCE NEE
2040	$154.83 [SOURCE NEEDED]	User-specified [SOURCE NEEDED]	Missing [SOURCE NEE
2050	$191.38 [SOURCE NEEDED]	User-specified [SOURCE NEEDED]	Missing [SOURCE NEE

Table 11: Grid electricity price trajectory – UNDOCUMENTED

5 Product Energy Benchmarks

Source File: `data/assumptions/product_benchmarks.csv`

CRITICAL DATA GAP: No sources provided for any of the 54 products.

5.1 NCC Products (Naphtha Cracker)

Product	Naphtha (GJ/t)	Elec (kWh/t)	LNG (GJ/t)	S
Ethylene	29.0 [SOURCE NEEDED]	85.4 [SOURCE NEEDED]	2.03 [SOURCE NEEDED]	N
Propylene	25.4 [SOURCE NEEDED]	191.3 [SOURCE NEEDED]	1.78 [SOURCE NEEDED]	N
Butadiene	29.0 [SOURCE NEEDED]	400.0 [SOURCE NEEDED]	1.88 [SOURCE NEEDED]	N

Table 12: NCC product energy benchmarks – ALL UNDOCUMENTED

5.2 BTX Products (BTX Plant)

Product	Naphtha (GJ/t)	Elec (kWh/t)	LNG (GJ/t)	Sou
Benzene	0.0 [SOURCE NEEDED]	36.6 [SOURCE NEEDED]	1.15 [SOURCE NEEDED]	Nor
Toluene	0.0 [SOURCE NEEDED]	26.8 [SOURCE NEEDED]	1.53 [SOURCE NEEDED]	Nor
Xylene	0.0 [SOURCE NEEDED]	763.1 [SOURCE NEEDED]	1.72 [SOURCE NEEDED]	Nor

Table 13: BTX product energy benchmarks – ALL UNDOCUMENTED

5.3 Polymer Products (Utility Process)

Product	Electricity (kWh/t)	Other Fuels	Source
HDPE	1,370.6 [SOURCE NEEDED]	0	None [SOURCE NEEDED]
PP	1,370.6 [SOURCE NEEDED]	0	None [SOURCE NEEDED]
LDPE	1,762.1 [SOURCE NEEDED]	0	None [SOURCE NEEDED]
PS	1,762.1 [SOURCE NEEDED]	0	None [SOURCE NEEDED]
PVC	1,488.0 [SOURCE NEEDED]	0	None [SOURCE NEEDED]

Table 14: Polymer product energy benchmarks – ALL UNDOCUMENTED

Note: Full list contains 54 products. All lack source citations.

Recommended Sources to Investigate:

- Korea Institute of Energy Research (KIER) benchmarks
- IEA Energy Technology Perspectives
- DECHEMA Technology Studies
- Company-specific technical documentation

6 Asset Valuation Parameters

Source File: `data/assumptions/asset_valuation_params.csv`

Process	CAPEX ($/ton)	Useful Life	Discount Rate
Naphtha Cracker	$1,500 [SOURCE NEEDED]	40 years [SOURCE NEEDED]	5% [SOURCE NE
BTX Plant	$800 [SOURCE NEEDED]	40 years [SOURCE NEEDED]	5% [SOURCE NE
Utility	$500 [SOURCE NEEDED]	30 years [SOURCE NEEDED]	5% [SOURCE NE
Other	$1,000 [SOURCE NEEDED]	30 years [SOURCE NEEDED]	5% [SOURCE NE

Table 15: Asset valuation parameters – ALL UNDOCUMENTED

Note: Discount rate in CSV is 5%, but code documentation mentions 8%. This inconsistency needs resolution.

7 Emission Reduction Targets

Source File: `data/scenarios/emission_targets.csv`

Year	Reduction Target	Source	Status
2025	0% (baseline) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing [SOURCE NE
2030	15% [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing [SOURCE NE
2035	24.5% [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing [SOURCE NE
2040	50% [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing [SOURCE NE
2045	75% [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing [SOURCE NE
2050	100% (Net Zero) [SOURCE NEEDED]	None [SOURCE NEEDED]	Missing [SOURCE NE

Table 16: Emission reduction pathway – ALL UNDOCUMENTED

Critical Question: Are these targets based on:

- Korea's Nationally Determined Contribution (NDC)?
- Industry-specific roadmap?
- Internal modeling assumption?

8 Scenario Definitions

Source File: `data/scenarios/scenario_definitions.csv`

Scenario	Production Pathway	NCC Technology	Sour
shaheen_ncc_h2	Shaheen (Growth) [SOURCE NEEDED]	NCC-H2	None
shaheen_ncc_elec	Shaheen (Growth) [SOURCE NEEDED]	NCC-Electricity	None
restructure_25pct_ncc_h2	25% Restructure [SOURCE NEEDED]	NCC-H2	None
restructure_25pct_ncc_elec	25% Restructure [SOURCE NEEDED]	NCC-Electricity	None
restructure_40pct_ncc_h2	40% Restructure [SOURCE NEEDED]	NCC-H2	None
restructure_40pct_ncc_elec	40% Restructure [SOURCE NEEDED]	NCC-Electricity	None

Table 17: Scenario definitions – rationale undocumented

Missing Documentation:

- Why 25% and 40% restructuring rates?

- What is the Shaheen pathway based on?
- Source for production growth/decline assumptions?

9 Data Gaps Summary

9.1 Critical Gaps (No Sources)

1. **Product Benchmarks** – 54 products × 8 energy columns = 432 values
2. **Fuel Prices** – 7 fuels × 26 years = 182 values (all constant)
3. **Asset Valuation** – 4 processes × 3 parameters = 12 values
4. **Emission Targets** – 6 milestone years
5. **Technology CAPEX** – All 5 technologies lack CAPEX sources

9.2 Partial Sources (Needs Specific Reference)

1. **Grid Emission Factor** – "Korea 2025" referenced but no document number
2. **H2 Demand Factor** – Derived from other values, methodology not cited
3. **Electricity Demand Factor** – Derived from other values, methodology not cited

9.3 Recommendations

1. Contact KPIA/KEEI for official energy benchmark data
2. Reference IEA WEO or EIA for fuel price projections
3. Cite Korea MOE official grid emission factor document
4. Document NDC or policy source for emission targets
5. Add methodology appendix for derived values (H2/elec demand factors)

Appendix: File Reference

Data Category	File Path
Technology Parameters	data/assumptions/technology_parameters.csv
Emission Factors	data/assumptions/emission_factors.csv
Product Benchmarks	data/assumptions/product_benchmarks.csv
Asset Valuation	data/assumptions/asset_valuation_params.csv
RE Prices	data/assumptions/prices/re_price_trajectory.csv
H2 Prices	data/assumptions/prices/h2_price_trajectory.csv
Fuel Prices	data/assumptions/prices/fuel_price_trajectory.csv
Grid EF	data/assumptions/prices/grid_emission_trajectory.csv
Grid Prices	data/assumptions/prices/grid_price_trajectory.csv
Emission Targets	data/scenarios/emission_targets.csv
Scenarios	data/scenarios/scenario_definitions.csv

Table 18: Data file reference

References

American Petroleum Institute (2021). Compendium of greenhouse gas emissions methodologies for the natural gas and oil industry. Technical report, API, Washington, DC.

BASF, SABIC, and Linde (2024). World's first electrically heated steam cracker furnace. Technical report, BASF Corporate Communications, Ludwigshafen. Joint press release on demonstration plant at BASF Verbund site.

Coolbrook (2024). Rotodynamic heater technology: Electrification of high-temperature industrial processes. Technical report, Coolbrook Oy, Helsinki.

Intergovernmental Panel on Climate Change (2019). 2019 refinement to the 2006 ipcc guidelines for national greenhouse gas inventories. Technical report, IPCC, Geneva.

International Energy Agency (2024). World energy outlook 2024. Technical report, IEA, Paris.

International Renewable Energy Agency (2024). Renewable power generation costs in 2023. Technical report, IRENA, Abu Dhabi.

Kosmadakis, G. (2020). Estimating the potential of industrial (high-temperature) heat pumps for decarbonizing eu industry. *Applied Thermal Engineering*, 171:115115.